

Homework 2

Problem 1. Prove the formula

1. $\binom{r}{r} + \binom{r+1}{r} + \binom{r+2}{r} + \cdots + \binom{n}{r} = \binom{n+1}{r+1}$
2. $\sum_{k=0}^n \binom{m+k-1}{k} = \binom{n+m}{n}$

Solution.

1. Use the equivalence $\binom{n+1}{r+1} = \binom{n}{r} + \binom{n}{r+1}$ iteratively.
2. Note that $\binom{m-1}{0} = \binom{m}{0} = 1$. The rest is just like above.

□

Problem 2. (a) Using Problem 1 for $r = 2$, calculate the sum $\sum_{i=2}^n i(i-1)$ and $\sum_{i=1}^n i^2$.

(b) Use (a) and Problem 1 for $r = 3$, calculate $\sum_{i=1}^n i^3$.

Solution.

1.

$$r = 2 : \quad \binom{2}{2} + \binom{3}{2} + \cdots + \binom{i}{2} + \cdots + \binom{n}{2} = \binom{n+1}{3}$$

$$\text{Thus } \frac{\sum_{i=2}^n i(i-1)}{2!} = \binom{n+1}{3} \therefore \sum_{i=2}^n i(i-1) = 2\binom{n+1}{3}$$

$$r = 1 : \quad \binom{1}{1} + \binom{2}{1} + \cdots + \binom{i}{1} + \cdots + \binom{n}{1} = \binom{n+1}{2}$$

$$\text{Thus } \therefore \sum_{i=1}^n i = \binom{n+1}{2}.$$

$$\text{Finally, } \sum_{i=1}^n i^2 = \sum_{i=1}^n (i(i-1) + i) = \sum_{i=1}^n i(i-1) + \sum_{i=1}^n i = \frac{n(n+1)(2n+1)}{6}.$$

2.

$$r = 3 : \quad \binom{3}{3} + \binom{4}{3} + \cdots + \binom{i}{3} + \cdots + \binom{n}{3} = \binom{n+1}{4}$$

$$\text{Thus } \frac{\sum_{i=3}^n i(i-1)(i-2)}{3!} = \binom{n+1}{4} \therefore \sum_{i=3}^n i^3 - 3i^2 + 2i = 6\binom{n+1}{4},$$

...

$$\text{The final result is } \binom{n+1}{2}^2.$$

□

Problem 3. How many functions $f : \{1, 2, \dots, n\} \rightarrow \{1, 2, \dots, n\}$ are there that are monotone; that is, for $i < j$ we have $f(i) \leq f(j)$?

Solution.

Set $k_i = f(i+1) - f(i)$, $i = 0, 1, \dots, n$, where we add $f(0) = 1$ and $f(n+1) = n$. Then the desired number is the number of nonnegative integer solutions to the equation $k_0 + k_1 + \dots + k_n = n - 1$.

Thus the final solution will be $\binom{2n-1}{n}$.

□